Interpreting a Chi-Square Result

Unit 3 · Topic 3.15 · TODAY'S PROBLEM

Suppose a city library takes an SRS of 240 of its 6,000 active adult cardholders. It records each person's chosen notification method and whether the person returned an item late in the previous six months.

Nia: "Email/Late has the largest raw gap, 13.75; I think that cell drives the statistic and the result shows email prevents late returns."

Omar: "Expected counts differ, so raw gaps may not settle cell influence. I would calculate contributions and examine how notification methods were obtained before interpreting."

Late returns

Method	Late	Not late	Total
Email	15	85	100
Text	24	56	80
None	30	30	60
Total	69	171	240

FIRST TAKE (5 min, on your sheet)

Can this study show that choosing email causes fewer late returns? Give one reason from how the study was done.

DOK 1 **(a)** State the degrees of freedom, excluding the totals row and column.

DOK 2 **(b)** Expected counts are 28.75 for Email/Late and 17.25 for None/Late. Calculate these two contributions using $(O - E)^2/E$.

DOK 3 **(c)** Technology gives $\chi^2 = 22.5172$ and $p = 0.0000129$. In 3–4 sentences, judge the accounts: compare the two contributions, use $\alpha = 0.05$ for a population conclusion, and explain why the study does not show prevention.

Optional review: follow-along (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.